Kernix 0.9.0 / Performance brief

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

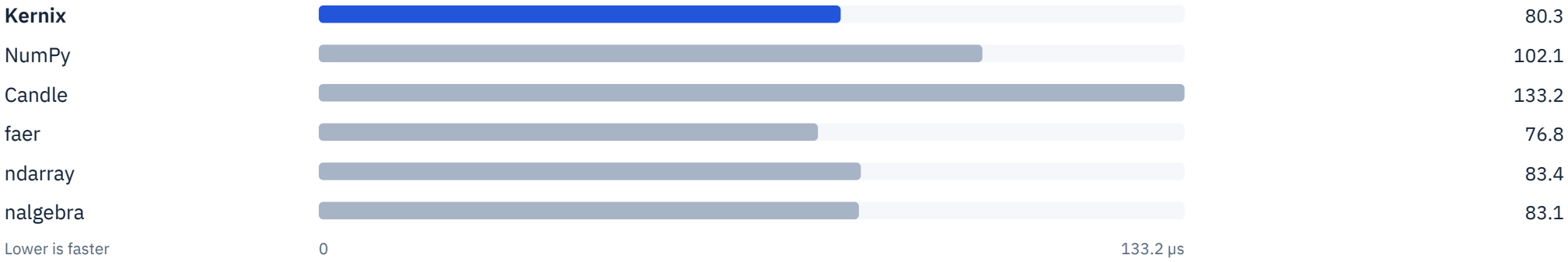
25 workloads

100 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

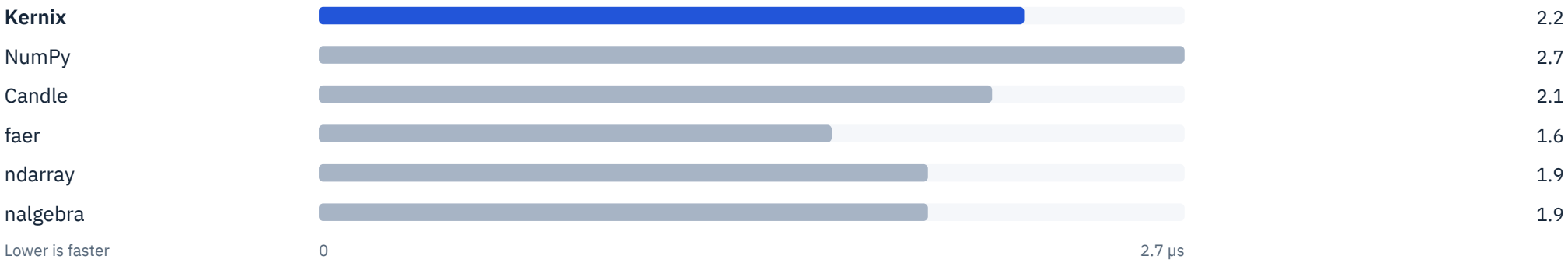
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



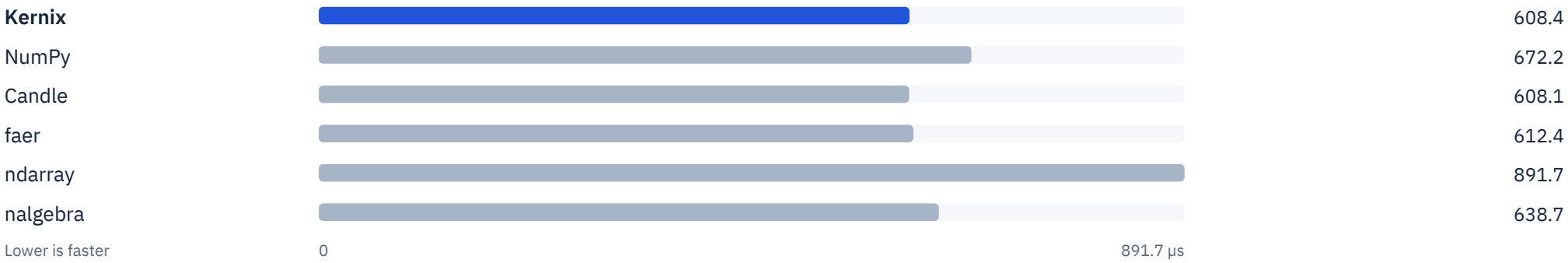
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



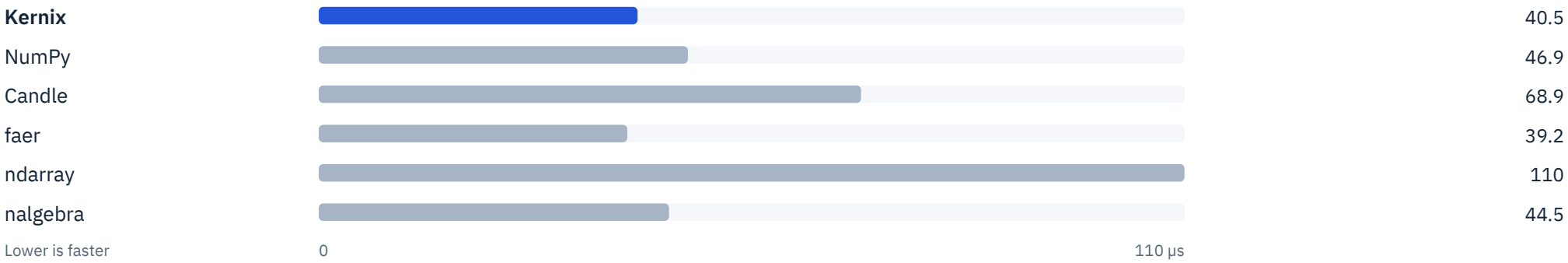
Kernix 0.9.0 / Matrix multiplication

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

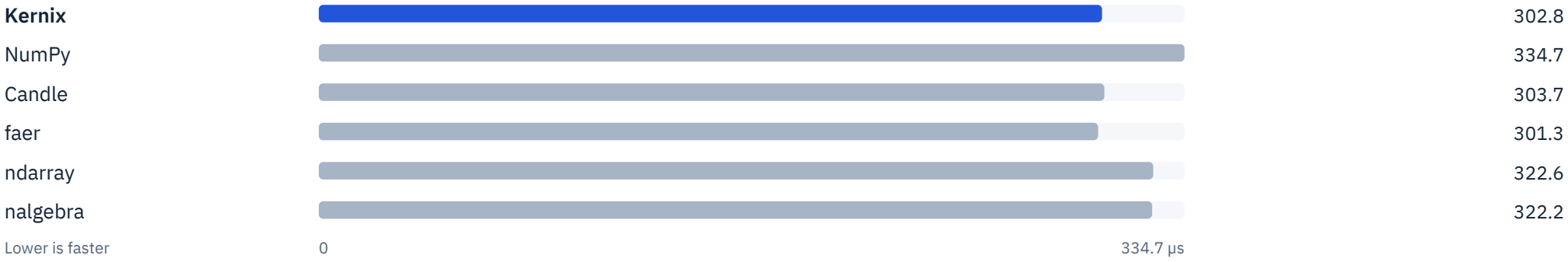
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.9.0 / Matrix multiplication

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

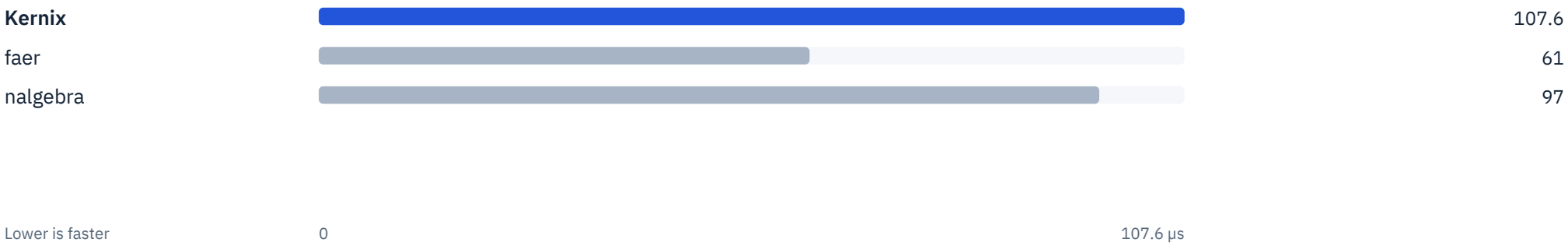


Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

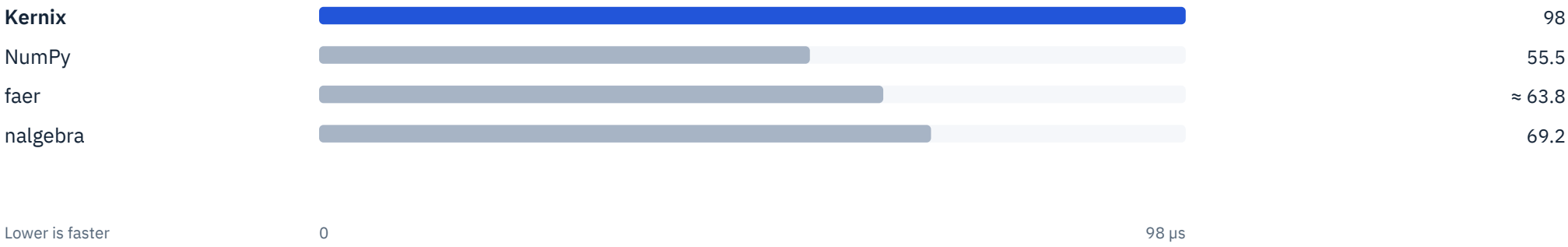


Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

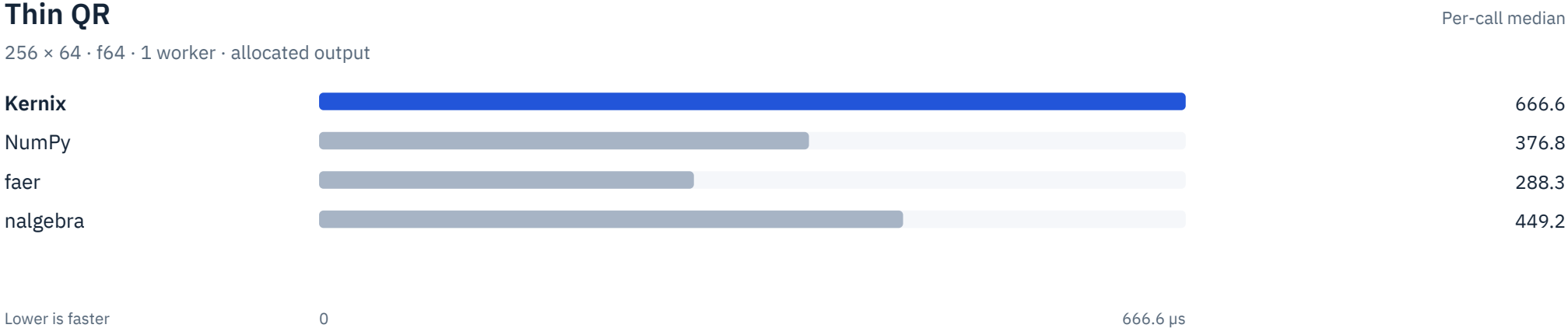
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



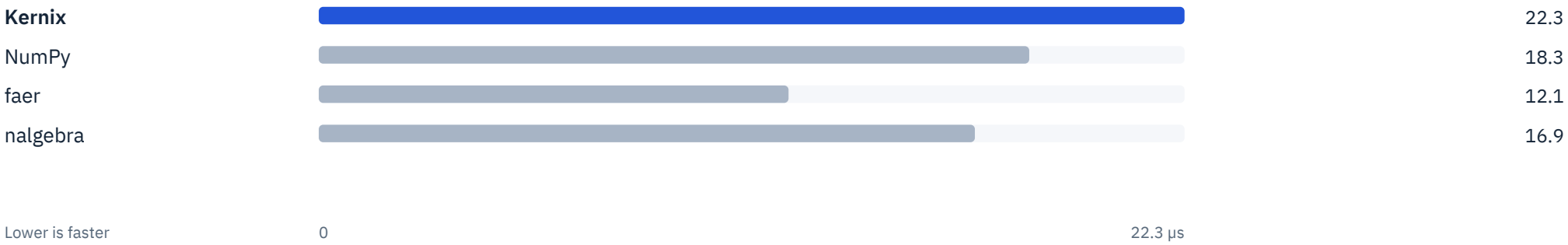
Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

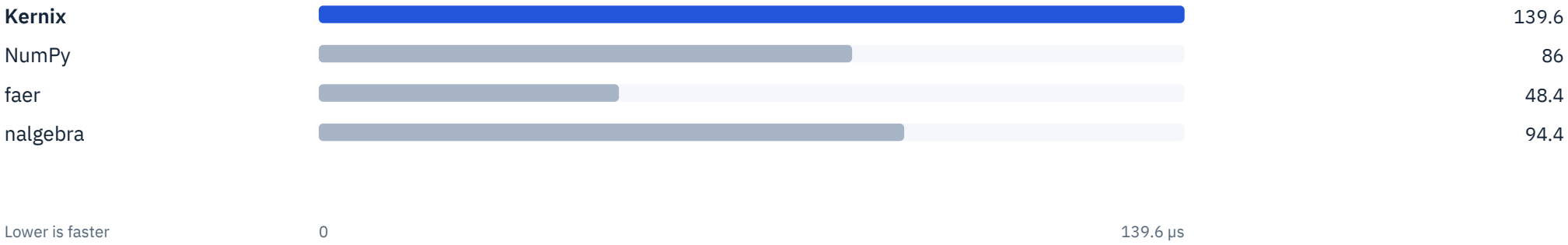
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



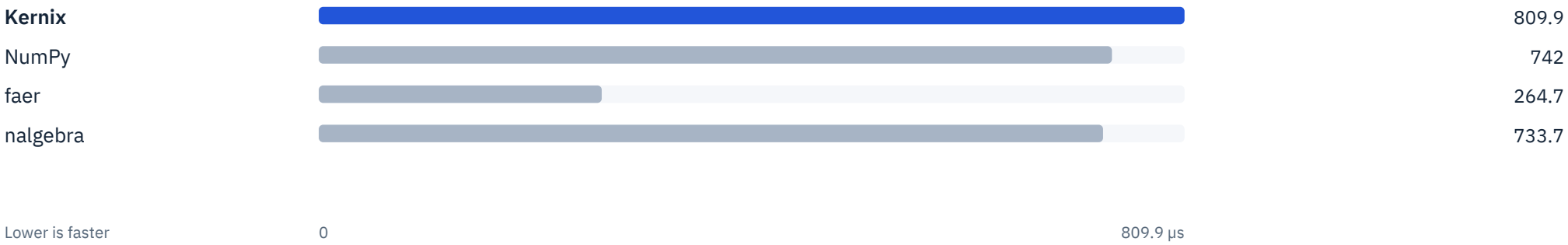
Kernix 0.9.0 / Decompositions & solvers

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.9.0 / Training & autodiff

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

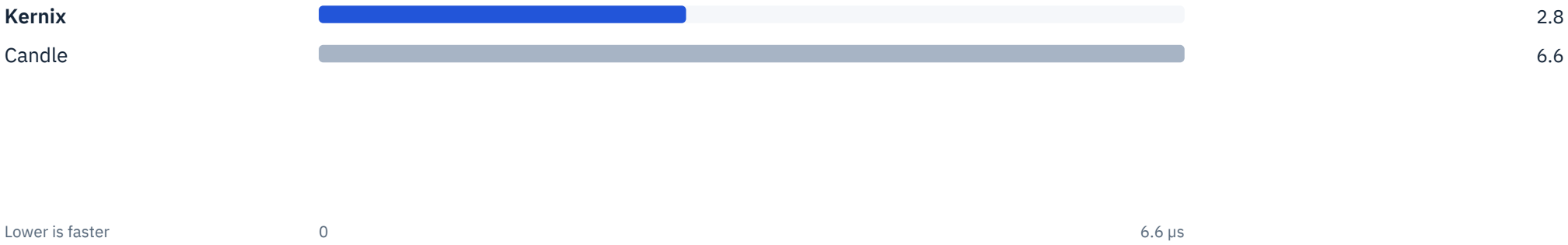
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.9.0 / Training & autodiff

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

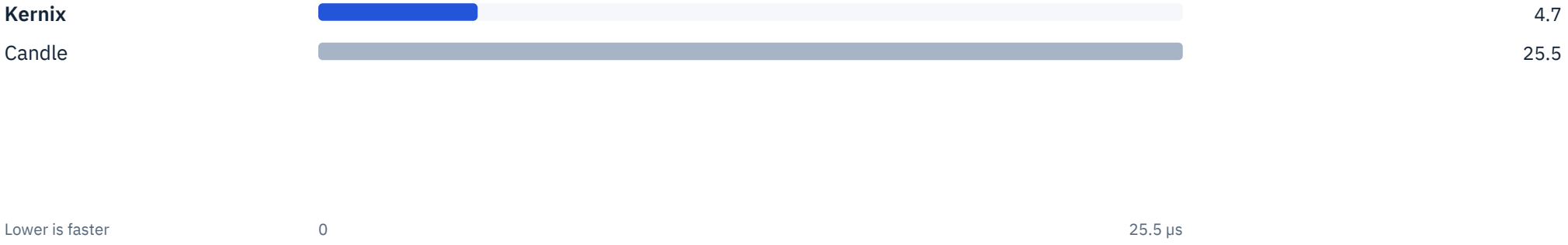
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.9.0 / Training & autodiff

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

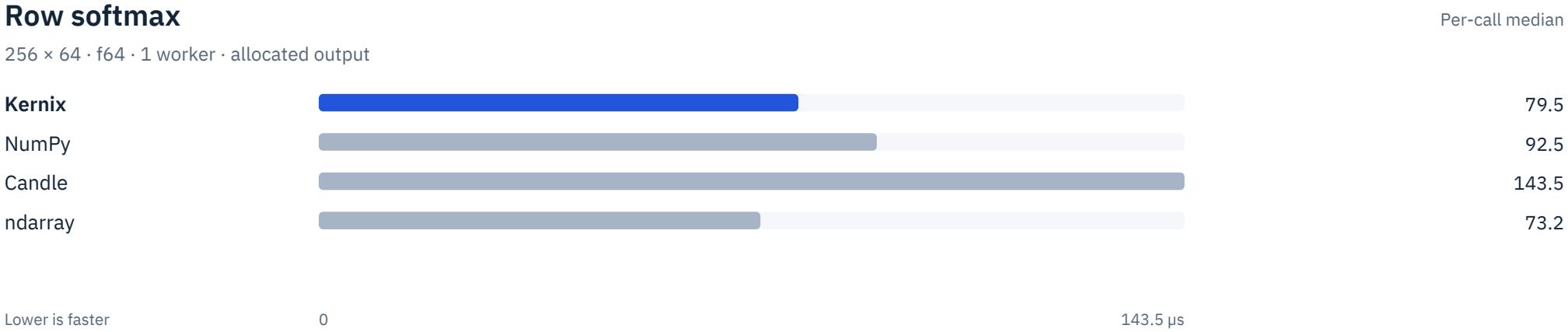


Kernix 0.9.0 / Statistics & activations

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



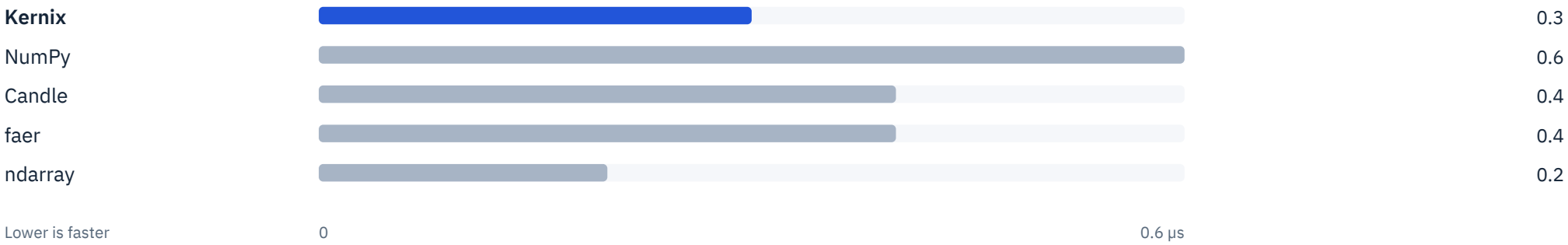
Kernix 0.9.0 / Tensor construction

AMD EPYC 9V74 80-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2200	2.2
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1600	1.6
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1900	1.9
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2100	2.1
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2700	2.7
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1900	1.9
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	10800	10.8
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	50300	50.3
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	11400	11.4
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	50900	50.9

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10600	10.6
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	11800	11.8
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	76800	76.8
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	102100	102.1
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	83100	83.1
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	83400	83.4
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	80300	80.3
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	133200	133.2
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	638700	638.7
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	672200	672.2

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	608400	608.4
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	608100	608.1
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	612400	612.4
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	891700	891.7
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	26100	26.1
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	6100	6.1
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6200	6.2
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	5700	5.7
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5500	5.5
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	26300	26.3

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	40500	40.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	46900	46.9
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	44500	44.5
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	68900	68.9
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	110000	110
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	39200	39.2
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	334700	334.7
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	301300	301.3
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	302800	302.8
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	322200	322.2

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	303700	303.7
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	322600	322.6
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3400	3.4
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2900	2.9
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2900	2.9
Column-pivoted QR / faer 96 × 32 · f64	61000	61
Column-pivoted QR / Kernix 96 × 32 · f64	107600	107.6
Column-pivoted QR / nalgebra 96 × 32 · f64	97000	97
Column-pivoted QR / nalgebra 256 × 64 · f64	863400	863.4
Column-pivoted QR / faer 256 × 64 · f64	320200	320.2

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / Kernix 256 × 64 · f64	752700	752.7
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	90800	90.8
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	85600	85.6
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	83800	83.8
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	153400	153.4
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	159000	159
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	138700	138.7
Thin QR / faer 96 × 32 · f64	63800.00000000001	63.80000000000001
Thin QR / NumPy 96 × 32 · f64	55500	55.5
Thin QR / nalgebra 96 × 32 · f64	69200	69.2

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 96 × 32 · f64	98000	98
Thin QR / faer 256 × 64 · f64	288300	288.3
Thin QR / Kernix 256 × 64 · f64	666600	666.6
Thin QR / nalgebra 256 × 64 · f64	449200	449.2
Thin QR / NumPy 256 × 64 · f64	376800	376.8
Cholesky factorization / Kernix 64 × 64 · f64	22300	22.3
Cholesky factorization / nalgebra 64 × 64 · f64	16900	16.9
Cholesky factorization / faer 64 × 64 · f64	12100	12.1
Cholesky factorization / NumPy 64 × 64 · f64	18300	18.3
Cholesky factorization / faer 128 × 128 · f64	48400	48.4

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	139600	139.6
Cholesky factorization / NumPy 128 × 128 · f64	86000	86
Cholesky factorization / nalgebra 128 × 128 · f64	94400	94.4
Cholesky factorization / nalgebra 256 × 256 · f64	733700	733.7
Cholesky factorization / faer 256 × 256 · f64	264700	264.7
Cholesky factorization / Kernix 256 × 256 · f64	809900	809.9
Cholesky factorization / NumPy 256 × 256 · f64	742000	742
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	5600	5.6
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	8000	8
Model inference / Kernix batched normalized/sliced MLP · f64	2800	2.8

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	6600	6.6
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3800	3.8
Model recording + forward / Candle batched normalized/sliced MLP · f64	8300	8.3
Model backward / Candle prepared graph → fresh gradients · f64	25500	25.5
Model backward / Kernix prepared graph → fresh gradients · f64	4700	4.7
Parameter update / Kernix transactional SGD, fixed gradients · f64	800	0.8
Row softmax / ndarray 256 × 64 · f64	73200	73.2
Row softmax / NumPy 256 × 64 · f64	92500	92.5
Row softmax / Candle 256 × 64 · f64	143500	143.5
Row softmax / Kernix 256 × 64 · f64	79500	79.5

Recorded precision / exact latency table

AMD EPYC 9V74 80-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	300	0.3
Create zeros / faer 64 × 64 · f32	400	0.4
Create zeros / ndarray 64 × 64 · f32	200	0.2
Create zeros / NumPy 64 × 64 · f32	600	0.6
Create zeros / Candle 64 × 64 · f32	400	0.4
Create ones / faer 64 × 64 · f32	400	0.4
Create ones / Kernix 64 × 64 · f32	400	0.4
Create ones / Candle 64 × 64 · f32	600	0.6
Create ones / NumPy 64 × 64 · f32	1600	1.6
Create ones / ndarray 64 × 64 · f32	300	0.3